

Econometrics Preliminary Exam, June 2025

There are **FOUR** questions. Answer each part of each question. All questions are weighted equally. Within each question, each part will receive equal weight in grading. You have 20 minutes to read the exam and then four hours to complete the exam.

1 Probability and statistics

Consider random variables X and Z , where $Z \sim \text{Bernoulli}(p)$ and X follows a mixture distribution of two normals with $X|Z = 0 \sim N(0, 1)$ and $X|Z = 1 \sim N(1, 4)$.

For your reference, the moment generating function (mgf) of a random variable $Y \sim N(\mu, \sigma^2)$ is given by $M(t) = e^{\mu t + \sigma^2 t^2 / 2}$ with derivatives

$$\begin{aligned}\frac{dM(t)}{dt} &= e^{\mu t + \sigma^2 t^2 / 2}(\mu + \sigma^2 t); \\ \frac{d^2 M(t)}{dt^2} &= e^{\mu t + \sigma^2 t^2 / 2}(\mu + \sigma^2 t)^2 + e^{\mu t + \sigma^2 t^2 / 2} \sigma^2.\end{aligned}$$

1. Use the mgf to derive the first two moments of a random variable $Y \sim N(\mu, \sigma^2)$.
2. Derive $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathbb{V}[X]$.
3. The probability density function (pdf) of a $N(\mu, \sigma^2)$ random variable is $\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$. Write down the conditional pdf of X conditional on $Z = 1$. Denote the conditional pdf by $f_{X|Z=1}(\cdot)$. Write down the marginal pdf of X . Denote the marginal pdf by $f_X(\cdot)$.
4. Use $\Phi(\cdot)$ to denote the cumulative distribution function of a standard normal random variable (i.e., $N(0, 1)$). Calculate $\mathbb{P}[Z = 1, X \leq x]$ and $\mathbb{E}[Z|X \leq x]$.

Suppose you observe a random sample $Z_1, \dots, Z_n$ drawn from the $\text{Bernoulli}(p)$ distribution of Z .

5. Derive your maximum likelihood estimator of p . Denote it by $\hat{p}$.
6. Derive the asymptotic distributions of $\sqrt{n}(\hat{p}^2 - p^2)$ as $n \rightarrow \infty$. Provide the formula for the standard error of the estimator $\hat{p}^2$.

Suppose you do not observe the random sample $Z_1, \dots, Z_n$. Instead, you observe a random sample $X_1, \dots, X_n$ drawn from the mixture normal distribution of X .

7. Construct a moment estimator of p using the first moment of X . Call your estimator $\hat{p}_{M,1}$. Show consistency of $\hat{p}_{M,1}$. Derive the asymptotic distribution of $\sqrt{n}(\hat{p}_{M,1} - p)$ as $n \rightarrow \infty$.
8. Construct a moment estimator of p using the second moment of X . Call your estimator $\hat{p}_{M,2}$. Compare the mean squared error (MSE) of $\hat{p}_{M,1}$ and $\hat{p}_{M,2}$.

2 Linear models

Suppose that you observe an i.i.d. sample (Y_i, X_i, Z_i) , $i = 1, \dots, n$, where Y_i , X_i and Z_i are scalar random variables. Let $Y_i = X_i\beta + \varepsilon_i$, $\mathbb{E}(\varepsilon_i) = \mathbb{E}(X_i) = \mathbb{E}(Z_i) = 0$, $\mathbb{E}(\varepsilon_i Z_i) = 0$. Consider an estimator

$$\tilde{\beta} = \frac{\sum_{i=1}^n Y_i Z_i}{\sum_{i=1}^n X_i Z_i}. \quad (1)$$

1. Is $\tilde{\beta}$ consistent for β ? If yes, prove the result. If not, state any additional assumptions required.
2. Is $\tilde{\beta}$ unbiased conditional on $\{(X_i, Z_i)\}_{i=1}^n$ for β in general? Explain why or why not. In your argument, you can use the special case where the representation $X_i = \rho Z_i + \alpha \varepsilon_i$ holds.
3. Provide sufficient conditions for consistency of the following estimator:

$$\hat{\beta} = \frac{\sum_{i=1}^n Y_i X_i}{\sum_{i=1}^n X_i^2}. \quad (2)$$

4. What are the names of $\hat{\beta}$ and $\tilde{\beta}$? Give examples of economic applications where one can use $\tilde{\beta}$, but not $\hat{\beta}$? State corresponding conditions on the data.
5. State any additional assumptions required for the asymptotic normality of the estimators. Provide a brief explanation for how these conditions are sufficient for the result. No need to provide a proof.
6. Derive the asymptotic variance of $\tilde{\beta}$ and $\hat{\beta}$.
7. Suppose that $\mathbb{E}(\varepsilon_i^2 | X_i, Z_i) = \sigma^2$ and sufficient conditions for the consistency of both estimators are satisfied. Provide a heuristic argument for the asymptotic efficiency comparison of the two estimators.

3 Heteroscedasticity and nonlinear models

- (a) Suppose we have i.i.d. observations $\{(Z_i, X_i) : i = 1, \dots, n\}$ such that $Z_i = X_i' \gamma + \varepsilon_i$, where $\mathbb{E}(\varepsilon_i | X_i) = 0$ and $\mathbb{E}(\varepsilon_i^2 | X_i) = \exp(X_i' \beta_0)$. Assume that β_0 is known.

1. Consider an estimator of γ

$$\hat{\gamma} = \left(\sum_{i=1}^n W_i X_i X_i' \right)^{-1} \sum_{i=1}^n W_i X_i Z_i \quad (3)$$

where $W_i = f(X_i)$ for some measurable function f . Is this estimator unbiased for γ ? State any additional assumptions required for the result.

2. Propose a choice of $f(\cdot)$ that minimizes the asymptotic variance of $\hat{\gamma}$.
 3. Propose a test statistic for the hypothesis $\gamma_1 = 0$, where γ_1 is the first component of the vector γ . State any additional assumptions.
- (b) Suppose we have i.i.d. observations $\{(Y_i, X_i) : i = 1, \dots, n\}$ with $\dim(Y_i) = 1$, $\dim(X_i) > 1$, $\mathbb{E}[Y_i^4] < \infty$ and $\|X_i\| \leq 1$.¹ Assume $\mathbb{E}[Y | X] = \exp(X' \beta_0)$ for some column vector β_0 . Suppose X satisfies no multicollinearity condition, i.e., $P(X' \beta_0 \neq X' \beta) > 0$ when $\beta \neq \beta_0$.

1. Prove the identification of the non-linear least squares estimator. In the proof, clarify where you use no multicollinearity condition. [Hint: Prove $\mathbb{E}[(Y - \exp(X' \beta))^2] = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2] + \mathbb{E}[(\mathbb{E}[Y | X] - \exp(X' \beta))^2]$ for any value β .]
2. Prove the non-linear least squares estimator is consistent. Make sure to state the sufficient conditions for the result.
3. Prove the non-linear least squares estimator is asymptotically normal. Make sure to state the sufficient conditions for the result.
4. Propose an asymptotic variance estimator for the non-linear least squares estimator. You do not need to prove anything for this question.

¹These assumptions are stronger than those often imposed in the literature.

4 Panel data

- (a) Suppose we have a panel dataset $\{(Y_{it}, X_{it}) : i = 1, \dots, N, t = 1, \dots, T\}$ with $\dim(Y_{it}) = 1$, $\dim(X_{it}) > 1$, $E[Y_{it}^4] < \infty$ and $E[\|X_{it}\|^4] < \infty$. Suppose the observations are i.i.d. across i . All asymptotic arguments in this part of the question pertain to the case where $N \rightarrow \infty$ holding T fixed.
1. Write down the regression model that corresponds to the fixed-effects estimator.
 2. Write down the moment equation corresponding to the fixed-effects estimator.
 3. Write down the objective function which the GMM estimator (based on the moment condition you derived in the above question) minimizes.
 4. The moment equation is just-identified in this case. Explain why the weight matrix does not affect the estimator.
 5. Prove the consistency for the fixed-effects estimator. Make sure to state the sufficient conditions for the result.
 6. Explain why the (pooled) OLS estimator is inconsistent. In your answer, derive the probability limit of the (pooled) OLS estimator.
- (b) We have clustered observations $\{(Y_{ig1}, Y_{ig2}, Y_{ig3}) : i = 1, \dots, N, g = 1, \dots, G\}$, where Y_{igt} denotes the outcome for unit i in cluster g during period $t = 1, 2, 3$. We assume the clusters are equal-sized and that $\{(Y_{ig1}, Y_{ig2}, Y_{ig3}) : i = 1, \dots, N\}$ are i.i.d. across g .² We consider the asymptotic framework with $G \rightarrow \infty$ while N is fixed. Consider the dynamic panel regression model

$$\begin{aligned} Y_{ig2} &= \alpha_0 Y_{ig1} + u_{ig} + \varepsilon_{ig1} \\ Y_{ig3} &= \alpha_0 Y_{ig2} + u_{ig} + \varepsilon_{ig2}. \end{aligned}$$

Assume

$$\mathbb{E}[Y_{ig1}\varepsilon_{ig2}] = 0, \mathbb{E}[Y_{ig1}\varepsilon_{ig3}] = 0, \text{ and } \mathbb{E}[Y_{ig2}\varepsilon_{ig3}] = 0.$$

1. Propose an estimator of α_0 . [Hint: Use Anderson-Hsiao estimator or Arellano-Bond estimator.]
2. Prove the consistency of the above estimator. Make sure to state the sufficient conditions for the result.
3. Prove the asymptotic normality of the above estimator. Make sure to state the sufficient conditions for the result.
4. Propose a cluster-robust asymptotic variance estimator.
5. Prove the consistency of the asymptotic variance estimator. Make sure to state the sufficient conditions for the result.

²These assumptions are stronger than those often imposed in the literature.

ARE/ECN 240B Reference Sheet

Probit	Logit
$\phi(u) = \frac{1}{\sqrt{2\pi}} \exp(-u^2/2)$	
$\Phi(u) = \int_{-\infty}^u \phi(t)dt$	$\Lambda(u) = (1 + \exp(-u))^{-1}$
$\frac{d}{du} \log \Phi(u) = \lambda(u)$ with $\lambda = \phi/\Phi$	$\frac{d}{du} \log \Lambda(u) = 1 - \Lambda(u)$
$\frac{d^2}{du^2} \log \Phi(u) = -\lambda(u)(u + \lambda(u)) < 0$	$\frac{d^2}{du^2} \log \Lambda(u) = -\Lambda(u)(1 - \Lambda(u)) < 0$
$\frac{d}{du} \phi(u) = -u\phi(u)$	
	$\frac{d}{du} \Lambda(u) = \Lambda(u)(1 - \Lambda(u))$
$\exists C > 0$ $\sup_u \lambda(u) < C$ $\sup_u \lambda(u)(u + \lambda(u)) < C$ $ \log \Phi(u) \leq C(1 + u) u $	

Matrix Calculus: (i) $\frac{\partial}{\partial x}(a'x) = \frac{\partial}{\partial x}(x'A) = a$. (ii) $\frac{\partial}{\partial x}(x'A) = A$ and $\frac{\partial}{\partial x'}(Ax) = A$. (iii) $\frac{\partial}{\partial x}(x'Ax) = (A + A')x$. (iv) $\frac{\partial^2}{\partial x \partial x'}(x'Ax) = A + A'$.

Theorem 2.1. If there is a function $Q_0(\theta)$ such that

1. $Q_0(\theta)$ is uniquely maximized at θ_0 ;
2. Θ is compact;
3. $Q_0(\theta)$ is continuous;
4. $\hat{Q}_n(\theta)$ converges uniformly in probability to $Q_0(\theta)$,

then $\hat{\theta} \xrightarrow{P} \theta_0$.

Lemma 2.4. If the data are i.i.d., Θ is compact, $a(z_i, \theta)$ is continuous at each $\theta \in \Theta$ with probability one, and there is $d(z)$ with $\|a(z, \theta)\| \leq d(z)$ for all $\theta \in \Theta$ and $\mathbb{E}[d(z)] < \infty$, then $\mathbb{E}[a(z, \theta)]$ is continuous and $\sup_{\theta \in \Theta} \left\| n^{-1} \sum_{i=1}^n a(z_i, \theta) - E[a(z, \theta)] \right\| \xrightarrow{P} 0$.

Theorem 2.7. If there is a function $Q_0(\theta)$ such that

1. $Q_0(\theta)$ is uniquely maximized at θ_0 ;
2. θ_0 is an element of the interior of a convex set Θ and $\hat{Q}_n(\theta)$ is concave; and
3. $\hat{Q}_n(\theta) \xrightarrow{P} Q_0(\theta)$ for all $\theta \in \Theta$,

then $\hat{\theta}_n$ exists with probability approaching one and $\hat{\theta}_n \xrightarrow{P} \theta_0$.

Theorem 3.1. Suppose that $\hat{\theta}$ satisfies eq. (1.1), $\hat{\theta} \xrightarrow{P} \theta_0$, and

1. $\theta_0 \in \text{interior}(\Theta)$;
2. $\hat{Q}_n(\theta)$ is twice continuously differentiable in a neighborhood $\mathcal{N}$ of θ_0 ;
3. $\sqrt{n} \nabla_{\theta} \hat{Q}_n(\theta_0) \xrightarrow{d} N(0, \Sigma)$;
4. there is $H(\theta)$ that is continuous at θ_0 and $\sup_{\theta \in \mathcal{N}} \|\nabla_{\theta\theta} \hat{Q}_n(\theta) - H(\theta)\| \xrightarrow{P} 0$;
5. $H = H(\theta_0)$ is nonsingular.

Then $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, H^{-1}\Sigma H^{-1})$.

Theorem 4.1. If the hypotheses of Theorem 3.1 are satisfied, $\hat{H} = \nabla_{\theta\theta} \hat{Q}_n(\hat{\theta})$, and $\hat{\Sigma} \xrightarrow{P} \Sigma$, then $\hat{H}^{-1} \hat{\Sigma} \hat{H}^{-1} \xrightarrow{P} H^{-1} \Sigma H^{-1}$.

Lemma 4.3. If z_i is i.i.d., $a(z, \theta)$ is continuous at θ_0 with probability one, and there is a neighborhood $\mathcal{N}$ of θ_0 such that $\mathbb{E}[\sup_{\theta \in \mathcal{N}} |a(z_i, \theta)|] < \infty$, then for any $\tilde{\theta} \xrightarrow{P} \theta_0$, $n^{-1} \sum_{i=1}^n a(z_i, \tilde{\theta}) \xrightarrow{P} E[a(z, \theta_0)]$.